



MATHIA

PRIJEMNI

Formule — Sve formule za prijemni iz matematike

I deo

Algebra, stepeni, koreni, logaritmi

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Uči s ljubavlju

Tvoja profesorka

I DEO Algebra, stepeni, koreni, logaritmi

Stepeni i koreni

$$a^m a^n = a^{m+n} \quad \frac{a^m}{a^n} = a^{m-n} \quad (a^m)^n = a^{mn} \quad a^{m/n} = \sqrt[n]{a^m} \quad a^{-n} = \frac{1}{a^n}$$

Logaritmi

$$\log_a(xy) = \log_a x + \log_a y \quad \log_a \frac{x}{y} = \log_a x - \log_a y \quad \log_a x^k = k \log_a x$$
$$\log_a x = \frac{\log_b x}{\log_b a} \quad a^{\log_a x} = x \quad \log_a a = 1, \quad \log_a 1 = 0$$

Skraćeno množenje i apsolutna vrednost

$$(a \pm b)^2 = a^2 \pm 2ab + b^2 \quad (a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$
$$a^2 - b^2 = (a - b)(a + b) \quad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$
$$|x| < c \iff -c < x < c \quad |x| > c \iff x < -c \vee x > c$$

II DEO Jednačine i nejednačine

Kvadratna jednačina

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}, \quad D = b^2 - 4ac$$
$$x_1 + x_2 = -\frac{b}{a} \quad x_1 x_2 = \frac{c}{a} \quad ax^2 + bx + c = a(x - x_1)(x - x_2)$$

Eksponencijalne i logaritamske

$$a^{f(x)} = a^{g(x)} \iff f(x) = g(x) \quad (a > 0, a \neq 1) \quad \log_a f = \log_a g \iff f = g > 0$$

III DEO Funkcije i nizovi

Linearna i kvadratna funkcija

$$y = kx + n, \quad k = \tan \alpha = \frac{y_2 - y_1}{x_2 - x_1} \quad y = ax^2 + bx + c: \quad x_T = -\frac{b}{2a}, \quad y_T = -\frac{D}{4a}$$

Aritmetički niz

$$a_n = a_1 + (n - 1)d \quad S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}[2a_1 + (n - 1)d]$$

Geometrijski niz

$$a_n = a_1 q^{n-1} \quad S_n = a_1 \frac{q^n - 1}{q - 1} \quad (q \neq 1) \quad |q| < 1 : S = \frac{a_1}{1 - q}$$

IV DEO Trigonometrija

Osnovni identiteti

$$\sin^2 \alpha + \cos^2 \alpha = 1 \quad \tan \alpha = \frac{\sin \alpha}{\cos \alpha} \quad 1 + \tan^2 \alpha = \frac{1}{\cos^2 \alpha}$$

Adicione formule i dvostruki ugao

$$\begin{aligned} \sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta & \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \sin 2\alpha &= 2 \sin \alpha \cos \alpha & \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha \end{aligned}$$

Sinusna i kosinusna teorema

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma} = 2R \quad a^2 = b^2 + c^2 - 2bc \cos \alpha$$

V DEO Geometrija i analitička geometrija

Površine ravnih figura

$$\begin{aligned} P_{\triangle} &= \frac{a h_a}{2} = \frac{1}{2} ab \sin \gamma & P_{\square} &= ab & P_{krug} &= r^2 \pi, \quad O = 2r\pi \\ \text{Heron: } P &= \sqrt{s(s-a)(s-b)(s-c)}, & s &= \frac{a+b+c}{2} \end{aligned}$$

Analitička geometrija u ravni

$$\begin{aligned} d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} & k &= \frac{y_2 - y_1}{x_2 - x_1} & y - y_1 &= k(x - x_1) \\ d(M, p) &= \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} & (x - p)^2 + (y - q)^2 &= r^2 \end{aligned}$$

VI DEO Kombinatorika i binomni obrazac

Prebrojavanje

$$P_n = n! \quad V_n^k = \frac{n!}{(n-k)!} \quad C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Binomni obrazac

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \quad \binom{n}{k} = \binom{n}{n-k}$$